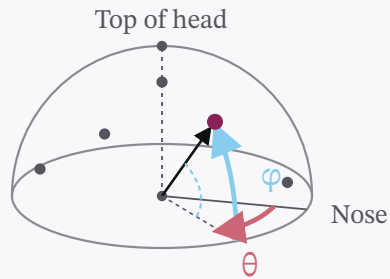


BrainLM-EEG: shared architecture and two proposed variants

BrainLM-EEG-RoPE

geometry-aware spatial encoding



azimuth θ , elevation ϕ
from the standard 10–20 montage

dims 0–63
rotated by θ

dims 64–127
rotated by ϕ

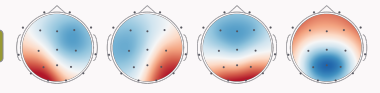
replaces channel embed
 $\text{CLS} \rightarrow \theta = \phi = 0$
rotates Q, K in every attention block

BrainLM-EEG-Microstate

secondary loss on the decoder output



mean-pooled over 19 channels
into 15 patch embeddings

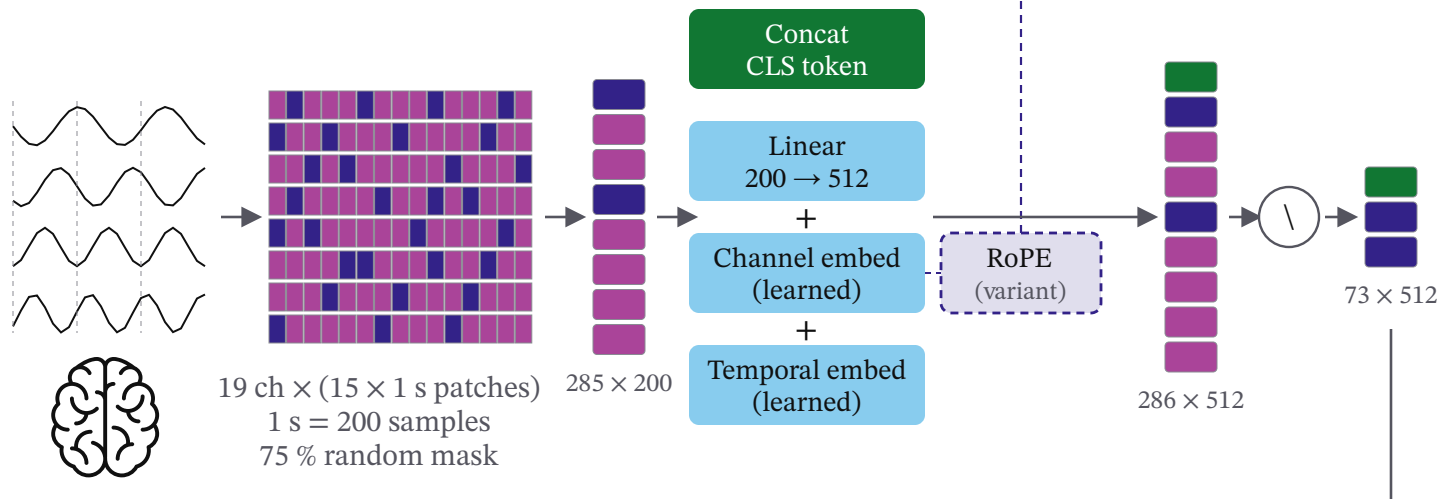


4 canonical maps, labels
from the LEMON train split

linear 512 $\rightarrow$ 32

8 sub-bins $\times$ 4 classes per 1 s patch

$$L = L_{\text{rec}} + \lambda \cdot L_{\text{aux}}, \lambda = 0.5$$



Colour stage

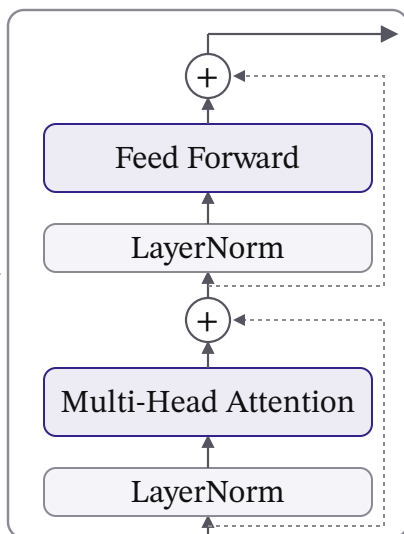
- Data
- Masked
- CLS token
- Encoding
- Prediction
- [MASK] token

Symbols

- $\backslash$ split into masked / unmasked
- $/$ join masked / unmasked
- $+$ element-wise addition

Encoder

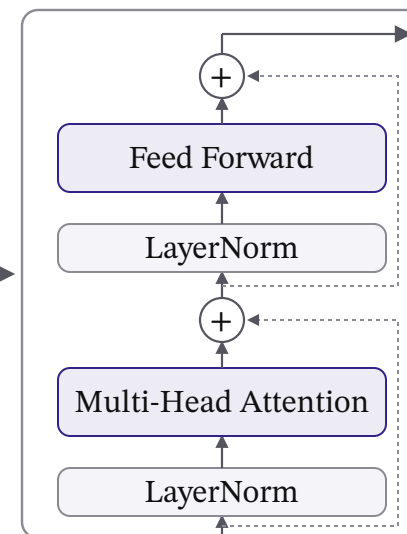
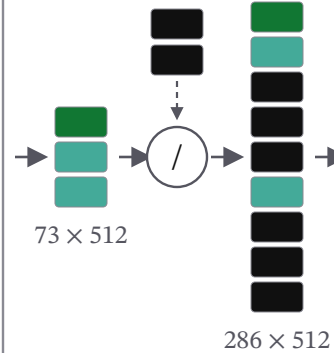
Transformer block $\times 4 \cdot \text{dim } 512 \cdot 4 \text{ heads}$



Decoder

Transformer block $\times 2 \cdot \text{dim } 512 \cdot 4 \text{ heads}$

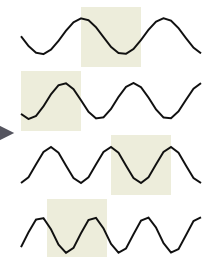
[MASK] $\times 213$



Microstate
(variant)

linear
512 $\rightarrow$ 200

285 patch tokens
(CLS dropped)



reconstruction loss on
masked (shaded) patches only